

## WORKED EXAMPLE A2.3

When a weight of 12.7 N was applied to a spring its length was 15.1 cm. When the force increased to 18.3 N, the length increased to 18.1 cm because the extension was proportional to the force.

- a Determine the spring constant.
- b Calculate the length of the spring when the force was 15.0 N.
- c Explain why it is impossible to be sure what the length of the spring would be if the force was 25 N.

**Answer**

a  $k = \frac{\Delta F}{\Delta x} = \frac{(18.3 - 12.7)}{(18.1 - 15.1)} = 1.87 \text{ N cm}^{-1}$ . Which is the same as  $187 \text{ N m}^{-1}$ .

- b Consider the extension from a length of 15.1 cm:

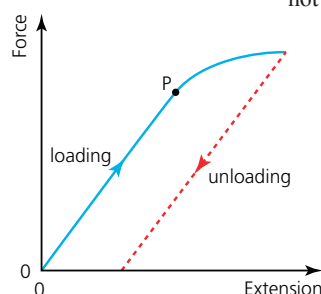
$$\Delta x = \frac{\Delta F}{k} = \frac{(15.0 - 12.7)}{1.87} = 1.23 \text{ cm}$$

So that, length =  $15.1 + 1.23 = 16.3 \text{ cm}$

- c The spring may have passed its limit of proportionality.

The results shown in Figure A2.19 were probably taken as the spring was *loaded* (as the weight was increased). If the extension is measured as the weight is *reduced* the results will be similar, but only if the elastic limit has not been exceeded.

The elastic limit of the spring is not shown on the graph, but it is often assumed to be close to, or the same as, the limit of proportionality. In other words, when a spring stretches, such as its extension is proportional to the force, we assume that it is behaving elastically. That may or may not be true for other materials.

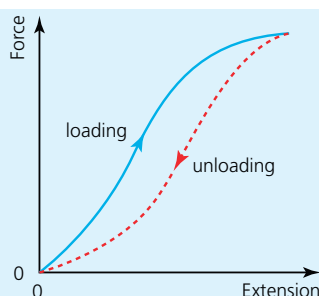


■ **Figure A2.20** Stretching a metal wire

Force–extension graphs and the concepts of elastic limits and ‘spring constants’ are not restricted to describing springs. They are widely used to represent the behaviour of many materials. Figure A2.20 shows a typical graph obtained when a metal wire is stretched and then the load is removed.

The force is proportional to the extension up until point P. During this time the particles in the metal are being pulled slightly further apart and we may assume that the metal is behaving elastically. But when the force is increased further, the wire begins to stretch more easily, the elastic limit is passed and a permanent deformation occurs. When the wire is unloaded the atoms move back closer together, so that the gradient of the graph is the same as for the loading graph, but the wire has a permanent deformation after all force has been removed.

- 11 A spring has a spring constant of  $125 \text{ N m}^{-1}$  and will become permanently deformed if its extension is greater than 20 cm.
  - a Assuming that it behaves elastically, what extension results from a tensile force of 18.0 N?
  - b What is the maximum force that should be used with this spring?
- 12 When a mass of 200 g was hung on a spring its length increased from 4.7 cm to 5.3 cm.
  - a Assuming that it obeyed Hooke’s law, what was its spring constant?
  - b The spring behaves elastically if the force does not exceed 10 N. What is the length of the spring with that force?
- 13 Figure A2.21 shows a force–extension graph for a piece of rubber which was first loaded, then unloaded.



■ **Figure A2.21** Stretching rubber

- a Does the rubber behave elastically? Explain your answer.
- b Does the rubber obey Hooke’s law under the circumstances shown by the graph? Explain your answer.

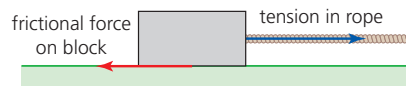
## Surface friction

### SYLLABUS CONTENT

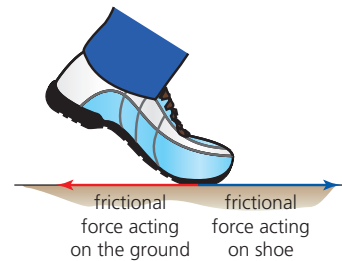
- Surface frictional force,  $F_f$ , acting in a direction parallel to the plane of contact between a body and a surface, on a stationary body as given by:  $F_f \leq \mu_s F_N$ , or a body in motion as given by:  $F_f = \mu_d F_N$ , where  $\mu_s$  and  $\mu_d$  are the coefficients of static and dynamic friction respectively.

◆ **Friction** Resistive forces opposing relative motion. Occurs between solid surfaces, but also with fluids. **Static friction** prevents movement, whereas **dynamic friction** occurs when there is already motion.

When we move an object over another surface (or try to move it), forces parallel to the surfaces will resist the movement. Collectively, these forces are known as surface **friction**. The causes of friction can be various, and it is well known that friction can often be difficult to analyse or predict. Figure A2.22 shows a typical simple frictional force diagram. (The frictional force acting on the ground is not shown.) The block is moving to the right and the frictional force is acting to the left.



■ **Figure A2.22** Frictional force on a block opposing its motion to the right

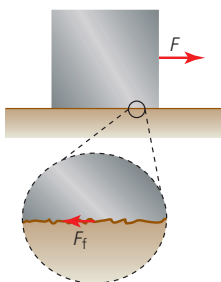


■ **Figure A2.23** We need friction to walk

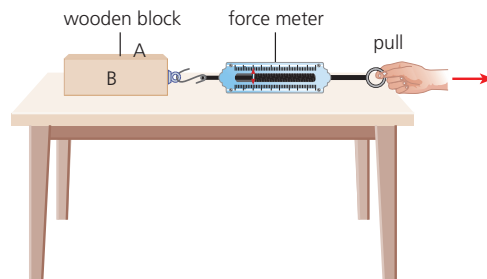
Friction is very useful: without friction we would not be able to walk. Similarly, a car's wheels would just spin on the same spot if there was no friction. Figure A2.23 explains why (the vertical forces are not shown). Because of friction, the shoe is able to push backwards, to the left, parallel to the ground, at the same time an equal frictional force pushes the shoe forward, to the right. (This is an example of Newton's third law of motion, which is discussed later in this topic.)

The roughness of both surfaces (see Figure A2.24) is certainly an important factor in producing friction: rougher surfaces generally increase friction, but this is not always true. For example, there may be considerable friction between very flat and smooth surfaces, like two sheets of glass. Friction can often be reduced by placing a lubricant, such as oil or water, between the surfaces.

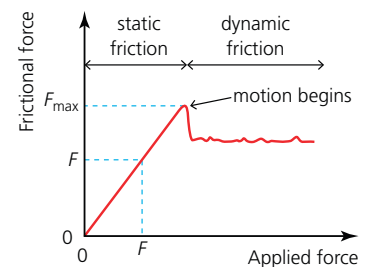
Figure A2.25 shows a basic laboratory investigation of the frictional forces between a wooden block and a horizontal table top.



■ **Figure A2.24** Even smooth surfaces have irregularities



■ **Figure A2.25** A simple experiment to measure frictional forces

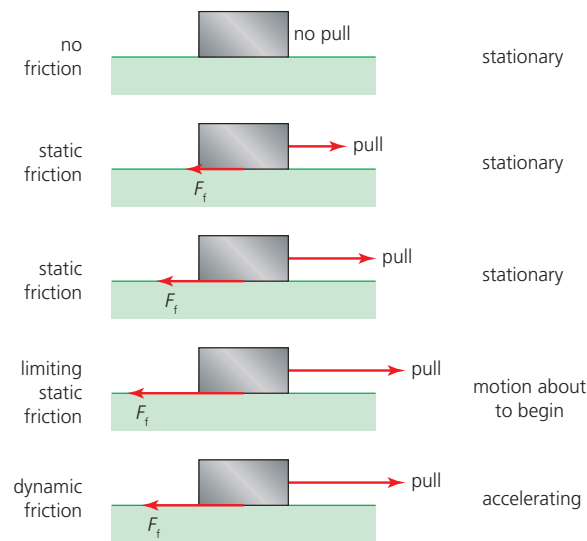


■ **Figure A2.26** Variation of friction with applied force

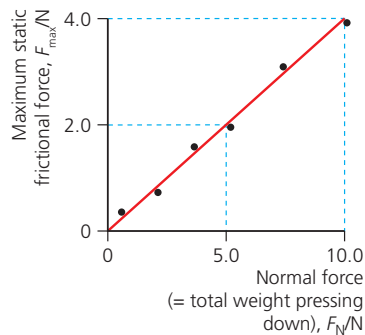
As the applied force (pull) is increased, the block will remain stationary until the force reaches a certain value,  $F_{\max}$ . The block then starts to move, but after that, a steady force, which is less than  $F_{\max}$ , will maintain a motion at constant speed. See Figure A2.26.

While the block is stationary (static) the force of friction adjusts, keeping equal to any applied force, but in the opposite direction. Under these circumstances the friction is called **static friction**. The size of the static friction force can increase from zero to a maximum value,  $F_{\max}$ . Once an object is moving, the reduced friction is called **dynamic friction**, and its value is approximately constant at different speeds.

Figure A2.27 illustrates how frictional forces can change as a pulling force is increased.



■ **Figure A2.27** How frictional forces change as the force applied increases



■ **Figure A2.28** Typical variation of maximum static frictional force with normal force (a similar pattern of results will be obtained for dynamic friction)



◆ **Coefficient of friction,  $\mu$**   
Constants used to represent the amount of friction between two different surfaces. Maybe static or dynamic.

The arrangement shown in Figure A2.25 can also be used to investigate how the *maximum* value of static friction depends on the force pushing the surfaces together: weights can be added on top of the block to increase the normal contact force,  $F_N$ . Figure A2.28 shows some typical results.

The graph shows that there is more static friction when there is a greater force pushing the surfaces together. In fact, frictional forces,  $F_f$ , are proportional to the normal contact forces,  $F_N$ . ( $F_f \propto F_N$ ) The constant of proportionality equals the gradient of the graph and is called the **coefficient of friction**,  $\mu$  (no units)

Just before motion begins:  $F_f = F_{\max} = \mu_s F_N$ , where  $\mu_s$  is the coefficient of static friction.

When there is no movement, static frictional force:  $F_f \leq \mu_s F_N$ .

Table A2.1 shows some typical values for the coefficient of static friction between different materials.

■ **Table A2.1** Approximate values for coefficients of static friction

| Materials   |                    | Approximate coefficients of static friction, $\mu_s$ |
|-------------|--------------------|------------------------------------------------------|
| steel       | ice                | 0.03                                                 |
| ski         | dry snow           | 0.04                                                 |
| Teflon™     | steel              | 0.05                                                 |
| graphite    | steel              | 0.1                                                  |
| wood        | concrete           | 0.3                                                  |
| wood        | metal              | 0.4                                                  |
| rubber tyre | grass              | 0.4                                                  |
| rubber tyre | road surface (wet) | 0.5                                                  |
| glass       | metal              | 0.6                                                  |
| rubber tyre | road surface (dry) | 0.8                                                  |
| steel       | steel              | 0.8                                                  |
| glass       | glass              | 0.9                                                  |
| skin        | metal              | 0.9                                                  |



When there is movement, dynamic frictional force,  $F_f = \mu_d F_N$ , where  $\mu_d$  is the coefficient of dynamic friction.

◆ **Constant** A number which is assumed to have the same numerical value under a specified range of circumstances.

◆ **Fundamental constants** Numbers which are assumed to have exactly the same numerical values under all circumstances and all times.

◆ **Coefficient** A multiplying constant placed before a variable, indicating a physical property.

### Tool 3: Mathematics

#### Applying general mathematics: constants

A number which is assumed to be **constant** always has the same value under the specified circumstances. For example, the spring constant described earlier in this topic represents the properties of a spring, but only up to its limit of proportionality. In Topic A.1, the acceleration due to gravity was assumed to be constant at  $9.8 \text{ m s}^{-2}$ , but only if we limit precision to 2 significant figures and only apply it to situations close to the Earth's surface.

However, there are a few constants which are believed to have exactly the same value in all locations and for all time. They are called the **fundamental constants**, or universal constants. Two examples are the speed of light and the charge on an electron.

In general, a **coefficient** is a number (usually a constant) placed before a variable in an algebraic expression. For example, in the expression  $5a - 2 = 8$ , the number 5 is described as a coefficient. In physics, a coefficient is used to characterize a physical process under certain specified conditions.

We have seen that: dynamic frictional force,  $F_f = \text{coefficient of dynamic friction} \times F_N$

Another example (which is not in the IB course): when a metal rod is heated it expands so that increase in length for each  $1^\circ\text{C}$  temperature rise  
 $= \text{coefficient of thermal expansion} \times \text{original length}.$

Objects also experience friction when they move through liquids and gases (fluids). This is discussed in the next section.



## ATL A2A: Research skills

### Using search engines and libraries effectively

#### Tyres and road safety

Much of road safety is dependent on the nature of road surfaces and the tyres on vehicles. Friction between the road and a vehicle provides the forces needed for any change of velocity – speeding up, slowing down, and changing direction. Smooth tyres will usually have the most friction in dry conditions, but when the roads are wet, ridges and grooves in the tyres are needed to disperse the water (Figure A2.29).

To make sure that road surfaces produce enough friction, they cannot be allowed to become too smooth and they may need to be resurfaced every few years. This is especially important on sharp corners and hills. Anything that gets between the tyres and the road surface – for example, oil, water, soil, ice and snow – is likely to affect friction and may have a significant effect on road safety. Increasing the area of tyres on a vehicle will change the pressure underneath them and this may alter the nature of the contact between the surfaces. For example, a farm tractor may have a problem about sinking into soft ground, and such a

situation is more complicated than simple friction between two surfaces. Vehicles that travel over soft ground need tyres with large areas to help avoid this problem.

Using a search engine, research online to find what materials are used in the construction of tyres and road surfaces to produce high coefficients of friction. Organize your data in a table, making sure to credit your sources using a recognized, standard method of referencing and citation.



Figure A2.29 Tread on a car tyre

### Common mistake

Many students expect that, if the block in Figure A2.25 was rotated so that side B was in contact with the table (instead of the side parallel to A), there would be more friction because of the greater area of contact. However, the frictional force will remain (approximately) the same, because if, for example, the area doubles, the force acting down on each  $\text{cm}^2$  will halve.

## WORKED EXAMPLE A2.4

- Determine the coefficient of friction for the two surfaces represented in the graph shown in Figure A2.28.
- Assuming the results were obtained for apparatus like that shown in Figure A2.25, calculate the minimum force that would be needed to move a block of total mass:
  - 200 g
  - 2000 g.
  - Suggest why the answer to part ii is unreliable.
- Estimate a value for the dynamic frictional force acting on a mass of 200 g with the same apparatus:
  - for movement at  $1.0 \text{ m s}^{-1}$
  - for movement at  $2.0 \text{ m s}^{-1}$ .

### Answer

- $\mu_s = \frac{F_{\text{max}}}{F_N} = \frac{4.0}{10.0} = 0.40$  (This is equal to the gradient of the graph.)
- $F_f = \mu_s F_N = \mu_s mg = 0.40 \times 0.200 \times 9.8 = 0.78 \text{ N}$
  - $0.40 \times 2.000 \times 9.8 = 7.8 \text{ N}$
  - Because the answer is extrapolated from well outside the range of experimental results shown on the graph.
- We would expect the dynamic frictional force to be a little less than the static frictional force, say about 0.6 N instead of 0.78 N.
  - The dynamic frictional force is usually assumed to be independent of speed, so the force would still be about 0.6 N at the greater speed.

Use data from Table A2.1 where necessary.

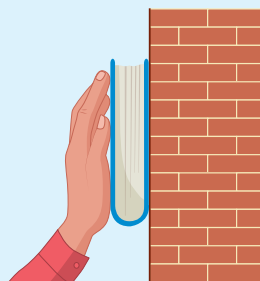
- 14** If dynamic friction is 85% of the maximum static friction, estimate the frictional force acting on the steel skates of a 47 kg ice-skater moving across the ice.
- 15** A 54 kg wooden box is on a horizontal concrete floor.
- a** Estimate the minimum force required to start it sliding sideways.
  - b** Suggest why your answer to part **a** may not be reliable.
  - c** If a force of 120 N keeps the box moving at a constant speed, what is the coefficient of dynamic friction?
  - d** What will happen to the box if the applied force increases above 120 N?
- 16**
- a** Predict the maximum frictional force possible between a dry road surface and each tyre of a stationary, 1500 kg four-wheeled family car.
  - b** Why will the force be less if the road is wet or icy?
  - c** Discuss how roads can be made safer under icy conditions.
- 17** Figure A2.30 shows the front of a Formula One racing car. Suggest how this design helps to increase the friction between the tyre and the race track.



■ **Figure A2.30** Front of a Formula One racing car

- 18** A book of mass 720 g is being held in place next to a vertical wall as shown in Figure A2.31.

- a** State the weight of the book.
- b** Suggest an approximate value for the coefficient of static friction between the book and the wall.
- c** Use your answer to part **b** to estimate the minimum force needed to keep the book stationary against the wall.



■ **Figure A2.31** Book being held next to a vertical wall

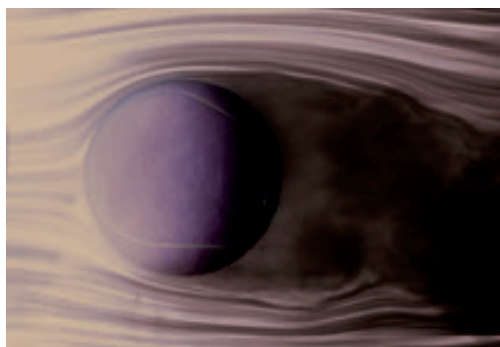
## Friction of objects with air and liquids

### SYLLABUS COVERAGE

- Viscous drag force,  $F_d$ , acting on a small sphere opposing its motion through a fluid as given by:  $F_d = 6\pi\eta rv$ , where  $\eta$  is the fluid viscosity,  $r$  is the radius of the sphere and  $v$  is the velocity of the sphere through the fluid.

Air resistance was briefly discussed in Topic A.1. The word *drag* is widely used to describe friction in air and liquids. We will use the symbol  $F_d$  for this type of force.





■ **Figure A2.32** Flow of air past a tennis ball in a wind tunnel.

There are a great number of applications of this subject, including moving vehicles, sports and falling objects. Wind tunnels are useful in the study of drag: the object is kept stationary while the speed of air flowing past it is varied. The flow of the air can be marked as shown in Figure A2.32.

Drag can be a complicated subject because the amount of drag experienced by an object moving through air, or a liquid, depends on many factors, including the object's size and shape, the nature of its surface, its speed  $v$ , and the nature of the fluid. Drag will also depend on the cross-sectional area of the object (perpendicular to its movement).

Typically, for small objects moving slowly  $F_d \propto v$ .

But for larger objects, moving more quickly,  $F_d \propto v^2$ .

## ■ Viscosity and Stokes's law

◆ **Viscosity** Resistance of a fluid to movement.

◆ **Viscous drag** The drag force acting on a moving object due to the viscosity of the fluid through which it is moving.

◆ **Turbulence** Flow of a fluid which is erratic and unpredictable.

◆ **Stokes's law** Equation for the viscous drag acting on a smooth, spherical object undergoing non-turbulent motion.

When an object moves through a fluid it has to push the fluid out of its path. A fluid's resistance to such movement is called its **viscosity**. Clearly, greater viscosity will tend to increase drag, and when this is the dominant factor, we refer to **viscous drag**.

Viscosity is given the symbol  $\eta$  (eta) and has the SI unit of Pa s ( $\text{kg m}^{-1} \text{s}^{-1}$ ). Some typical values at  $20^\circ\text{C}$  are given in Table A2.2. Viscosities of liquids can be very dependent on temperature.

■ **Table A2.2** Viscosities of some fluids

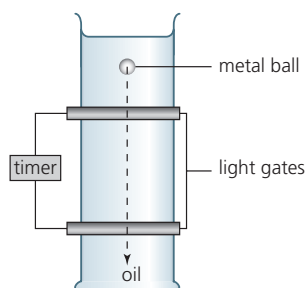
| Fluid             | Viscosity $\eta/\text{Pa s}$ |
|-------------------|------------------------------|
| 'heavy' oil       | 0.7                          |
| 'light' oil       | 0.1                          |
| water             | $1 \times 10^{-3}$           |
| human blood       | $4 \times 10^{-3}$           |
| gasoline (petrol) | $6 \times 10^{-4}$           |
| air               | $1.8 \times 10^{-5}$         |

In order to understand this further, we start by simplifying the situation, as is common in physics: by considering a smooth spherical object, of radius  $r$ , moving at a speed  $v$ , which is not great enough to cause **turbulence** (irregular movements) in the fluid.

Under these circumstances, the viscous drag,  $F_d$ , can be determined from the following equation (known as **Stokes's law**):



$$\text{viscous drag } F_d = 6\pi\eta rv$$



■ **Figure A2.33** Experiment to determine the viscosity of a liquid

Dropping small spheres through fluids is a widely used method for determining their viscosities and how they may depend on temperature. A method is shown in Figure A2.33, in which an electronic timer is started and stopped as the metal ball passes through the two light gates.

## Inquiry 1: Exploring and designing

### Designing

Look at the apparatus setup in Figure A2.33. Apply what you know about terminal speed (Topic A.1) and viscous fluid flow to design and explain a valid methodology for an experiment to obtain a single set of measurements. Include an explanation of:

- 1 why the metal ball is released such that it passes through some oil before reaching the first timing gate
- 2 why the tube should be as wide as possible.



■ **Figure A2.34** Forces on a sphere falling with terminal speed



If a sphere of mass  $m$  and radius  $r$  is moving with a constant terminal speed,  $v_t$ , then the upwards and downwards forces on it are balanced, as shown in Figure A2.34.

viscous drag,  $F_d$  + buoyancy force,  $F_b$  = weight,  $mg$ :

$$6\pi\eta rv + \rho Vg = mg$$

but:

$$V = \frac{4}{3}\pi r^3$$

so:

$$6\pi\eta rv + \frac{4}{3}\rho g\pi r^3 = mg$$

If the mass and radius of the sphere are measured and the terminal speed determined as shown in Figure A2.33, then this equation can be used to determine a value for the viscosity of the liquid, assuming that its density is known.

### Inquiry 3: Concluding and evaluating

#### Evaluating

The experimental determination of a viscosity discussed above involved just one set of measurements and a calculation.

Explain improvements to increase the accuracy of the determination of the viscosity of a liquid by collecting sufficient data to enable a graph of the results to be drawn.

### WORKED EXAMPLE A2.5

Calculate the force of viscous drag on a sphere of radius 1.0 mm moving at 1.0 cm s<sup>-1</sup> through 'heavy' oil.

**Answer**

$$F_d = 6\pi \times \eta \times r \times v = 6 \times 3.14 \times 0.7 \times (1.0 \times 10^{-3}) \times (1.0 \times 10^{-2}) = 1.3 \times 10^{-4} \text{ N}$$

- 19** The air resistance acting on a car moving at 5.0 m s<sup>-1</sup> was 120 N. Assuming that this force was proportional to the speed squared, what was the air resistance when the car's speed increased to:

**a** 10 m s<sup>-1</sup>      **b** 15 m s<sup>-1</sup>?

- 20** Show that the units of viscosity are Pa s.

- 21** Calculate the viscous drag force acting on a small metal sphere of radius 1.3 mm falling through oil of viscosity 0.43 Pa s at a speed of 7.6 cm s<sup>-1</sup>.

- 22** A drop of water in a cloud had a mass of 0.52 g and radius of 0.50 mm (and volume of 0.52 mm<sup>3</sup>).

**a** Assuming that the density of the surrounding air is 1.3 kg m<sup>-3</sup>, calculate and compare the size of the three

forces acting on the drop if it has just started to fall with a speed of 5.0 cm s<sup>-1</sup>.

- b** Draw an annotated diagram to display your answers.  
**c** Determine the subsequent movement of the drop.

- 23** In an experiment similar to that shown in Figure A2.33, a sphere of radius 8.9 mm and mass 3.1 g reached a terminal speed of 7.6 cm s<sup>-1</sup> when falling through an oil of density 842 kg m<sup>-3</sup>. Determine a value for the viscosity of the liquid.

- 24** Use the internet to find out how the design of golf balls reduces drag forces in flight. Write a 100 word summary of your findings.